

Multiple Choice

1. **Answer:** B

Options: A) 6; B) 32; C) 30; D) 12; E) 36

Note: Correct option: B - 32

2. **Answer:** E

Options: A) conducting the survey at one shoe store; B) conducting the survey online targeting only the mall's registered customers; C) conducting the survey at all clothing stores; D) conducting the survey only on weekends; E) conducting the survey at the entrance to the mall

Note: Correct option: E - conducting the survey at the entrance to the mall

3. **Answer:** B

Options: A) 85.0; B) 60.0; C) 65.0; D) 45.0; E) 55.0

Note: Correct option: B - 60.0

4. **Answer:** A

Options: A) 13 in.; B) 10 in.; C) 11 in.; D) 45.0 in.; E) 15 in.

Note: Correct option: A - 13 in.

5. **Answer:** A

Options: A) 3.5; B) 1; C) 0.25; D) 0.5; E) 2.5

Note: Correct option: A - 3.5

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '6'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '240'.

Long Form Response

11. **Answer:** We begin by factoring 336. $336 = 2^4 \cdot 3 \cdot 7$. Because we are looking for phone numbers, we want four single digits that will multiply to equal 336. Notice that 7 cannot be multiplied by anything, because $7 \cdot 2$ is 14, which is already two digits. So, one of our digits is necessarily 7. The factor 3 can be multiplied by at most 2, and the highest power of 2 that we can have is $2^3 = 8$. Using these observations, it is fairly simple to come up with the following list of groups of digits whose product is 336 :

&1, 6, 7, 8

&2, 4, 6, 7

&2, 3, 7, 8

&3, 4, 4, 7 For the first three groups, there are $4! = 24$ possible rearrangements of the digits. For the last group, 4 is repeated twice, so we must divide by 2 to avoid overcounting, so there are $\frac{4!}{2} = 12$ possible rearrangements of the digits. Thus, there are $3 \cdot 24 + 12 = 84$ possible phone numbers that can be constructed to have this property.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** Since $\angle ABP = 90^\circ$, $\triangle ABP$ is a right-angled triangle. By the Pythagorean Theorem, $BP^2 = AP^2 - AB^2 = 20^2 - 16^2 = 144$ and so $BP = 12$, since $BP > 0$. Since $\angle QTP = 90^\circ$, $\triangle QTP$ is a right-angled triangle with $PT = 12$. Since $PT = BP = 12$, then by the Pythagorean Theorem,

$$QT^2 = QP^2 - PT^2 = 15^2 - 12^2 = 81$$

and so $QT = 9$, since $QT > 0$. Our final answer is then 12, 9\$.

Note: Students should show all steps in their mathematical solution.

End of Answer Key